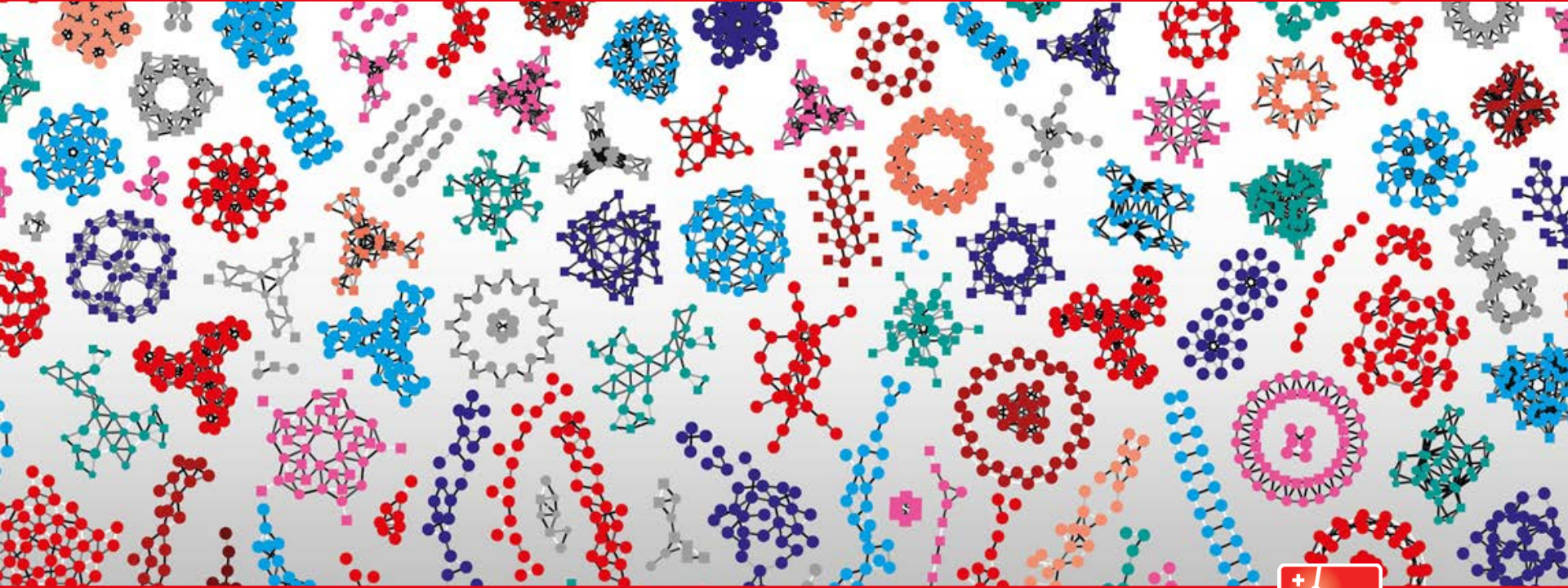




Swiss Institute of
Bioinformatics

Advanced R

August 2026

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Introduction

**What do you expect
from this course ?**

Welcome

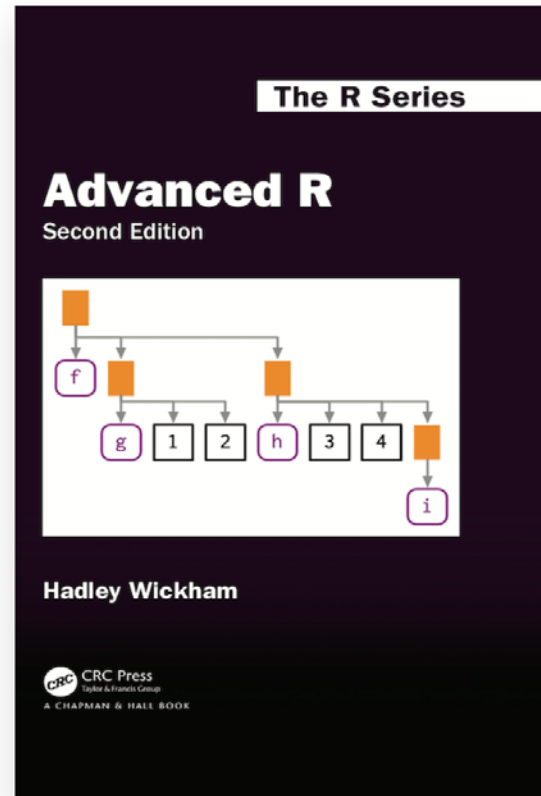
This is the website for 2nd edition of “**Advanced R**”, a book in Chapman & Hall’s R Series. The book is designed primarily for R users who want to improve their programming skills and understanding of the language. It should also be useful for programmers coming to R from other languages, as help you to understand why R works the way it does.

If you’re looking for the 1st edition, you can find it at <http://adv-r.had.co.nz/>.

License



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On this page

Welcome

License

Other books

[View source](#)

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Are these valid R commands ?

If yes, what do they do ?

If no, why ?

c=c (c=c)

c=c (c="c")

**How would you describe
"good programming" ?**

Reproducible research

**The
Economist**

OCTOBER 19TH-25TH 2013

Economist.com

Washington's lawyer surplus

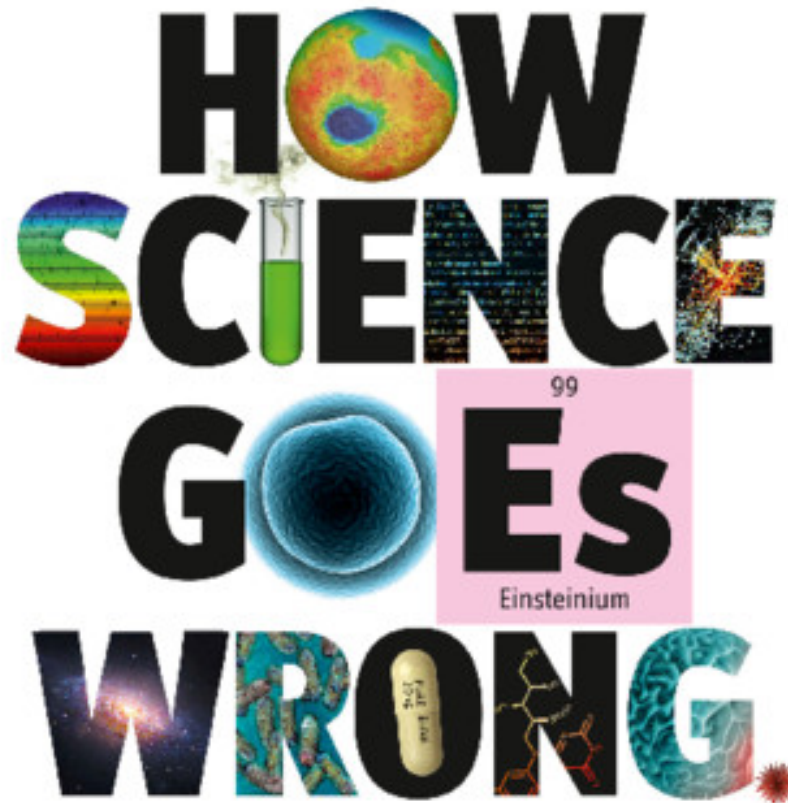
How to do a nuclear deal with Iran

Investment tips from Nobel economists

Junk bonds are back

The meaning of Sachin Tendulkar

HOW SCIENCE GOEs WRONG



THIS WEEK

EDITORIALS

FOOD Fertilizer is the best way to feed Africa right now **p.510**



WORLD VIEW Teach young scientists how to manage their labs **p.511**

VENICE City of Water edges closer to becoming a city under water **p.512**

Must try harder

Too many sloppy mistakes are creeping into scientific papers. Lab heads must look more rigorously at the data — and at themselves.

Science: Branch of knowledge or study dealing with a body of facts or truths systematically arranged. So says the dictionary. But, as

for the first time only when problems in published studies are reported. In private, scientists who run labs in even the most prestigious uni-

Must try harder

Too many sloppy mistakes are creeping into scientific papers. Lab heads must look more rigorously at the data — and at themselves.

technical reasons why basic research findings do not stand up in clinical work. But the overall impression the article leaves is of insufficient thoroughness in the way that too many researchers present their data.

The finding resonates with a growing sense of unease among specialist editors on this journal, and not just in the field of oncology. Across the life sciences, handling corrections that have arisen from avoidable errors in manuscripts has become an uncomfortable part of the publishing process.

The evidence is largely anecdotal. So here are the anecdotes: unrelated data panels; missing references; incorrect controls; undeclared cosmetic adjustments to figures; duplications; reserve figures and dummy text included; inaccurate and incomplete methods; and improper use of statistics — the failure to understand the difference between technical replicates and independent experiments, for example.

It is usually the case that original data can be produced, mistakes corrected, and the findings of the corrected research paper still stand. At the very least, however, there is too little attention paid and too many corrections, which reflect unacceptable shoddiness in laboratories that risks damaging trust in the science that they, and others, produce.

The situation throws up many questions. Here are three of them. Who is responsible? Why is it happening? How can it be stopped?

The principal investigators (PIs) of any lab from which the work originates, especially if their names are on the paper, have an absolute and unavoidable responsibility to ensure the quality of the data from their labs, even if the main work is done by experienced postdocs. Officially, postdocs and graduate students are still in training, and it is the PI's job to make sure they are properly trained — in statistics and appropriate image editing, for a start. It is unacceptable for lab heads — who are happy to take the credit for good work — to look at raw data

from the original paper — as the Comment authors recommend.

There should also be increased scope to publish fuller results from an experiment, and subsequent negative or positive corroborations. There is an opportunity here for 'minimum threshold' journals, such as *PLoS ONE* and *Scientific Reports*. Editors and referees cannot be expected to divine when only positive data are included and inconvenient results left out, but journals should encourage online presentation of the complete picture. And scientists should offer it. The complete picture is, after all, what this science of ours strives to provide. ■

Nature editorial: "Must try harder"

Glenn Begley and Lee Ellis analyze the low number of cancer-research studies that have been converted into clinical success, and conclude that **a major factor is the overall poor quality of published preclinical data.**

[...]

The overall impression the article leaves is of *insufficient thoroughness in the way that too many researchers present their data.*

Reproducible Research:

“Research is reproducible if it can be reproduced by others”

Reproducible research

Of course, rerunning an experiment will give different results—an observation that gave rise to the development of statistics as a discipline.

Our focus here is “reproducible research” (RR) in the sense of reproducing conclusions from a single experiment based on the measurements from that experiment.



A **complete description** of the data and the analysis of that data — including computer programs — so the **results can be exactly reproduced** by others.

How complicated is it ?

Observing many researchers using the same data and hypothesis reveals a hidden universe of uncertainty

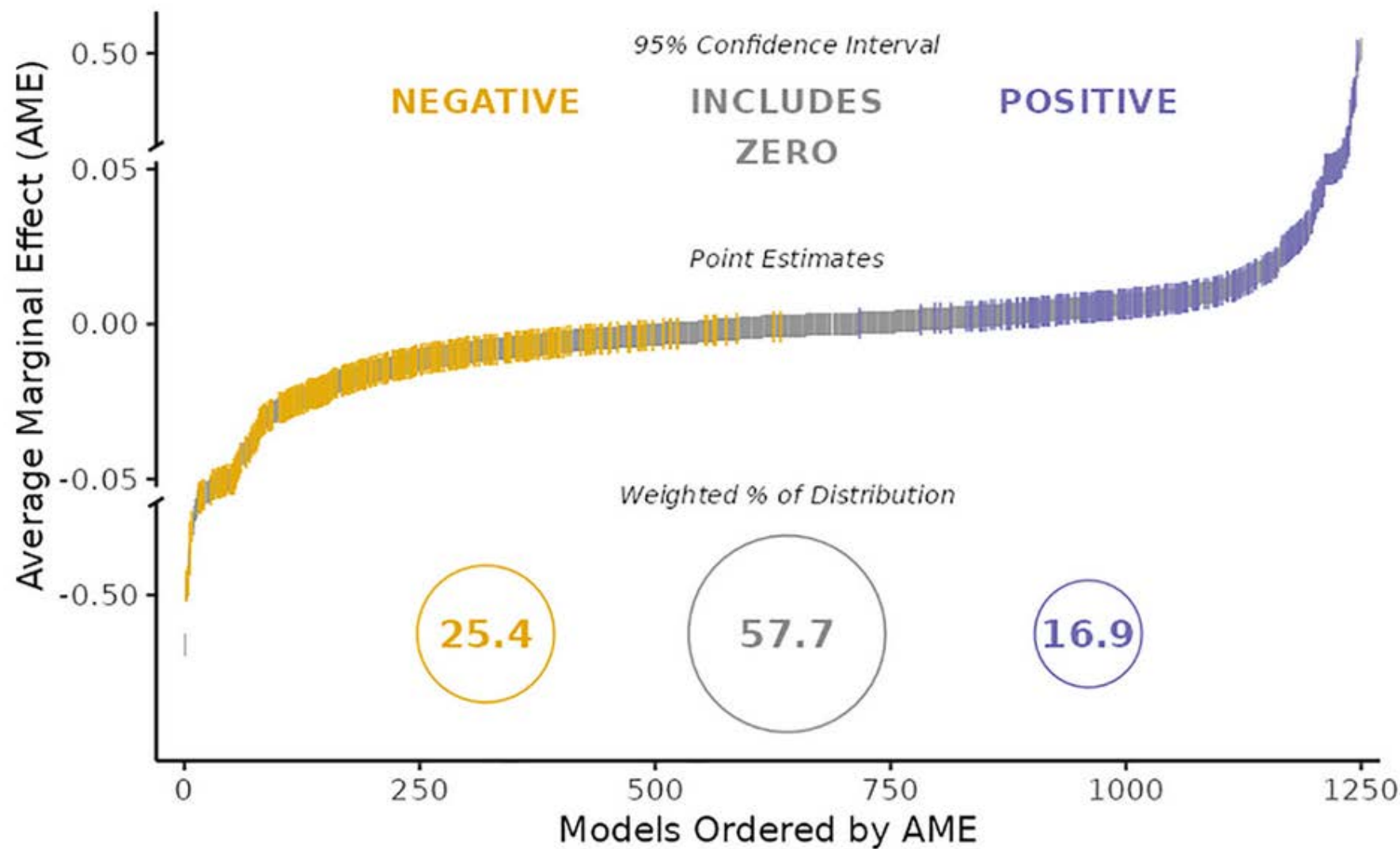
[Nate Breznau](#)  , [Eike Mark Rinke](#) , [Alexander Wuttke](#) ,  +162, and [Tomasz Żółtak](#)  [Authors Info & Affiliations](#)

Edited by Douglas Massey, Princeton University, Princeton, NJ; received March 6, 2022; accepted August 22, 2022

October 28, 2022 | 119 (44) e2203150119 | <https://doi.org/10.1073/pnas.2203150119>

We coordinated 161 researchers in 73 research teams and observed their research decisions as they used the same data to independently test the same prominent social science hypothesis.

Fig. 1.



DERIVING CHEMOSENSITIVITY FROM CELL LINES: FORENSIC BIOINFORMATICS AND REPRODUCIBLE RESEARCH IN HIGH-THROUGHPUT BIOLOGY

BY KEITH A. BAGGERLY^{*} AND KEVIN R. COOMBES[†]

U.T. M.D. Anderson Cancer Center

High-throughput biological assays such as microarrays let us ask very detailed questions about how diseases operate, and promise to let us personalize therapy. Data processing, however, is often not described well enough to allow for exact reproduction of the results, leading to exercises in “forensic bioinformatics” where aspects of raw data and reported results are used to infer what methods must have been employed. Unfortunately, poor documentation can shift from an inconvenience to an active danger when it obscures not just methods but errors. In this report, we examine several related papers purporting to use microarray-based signatures of drug sensitivity derived from cell lines to predict patient response. Patients in clinical trials are currently being allocated to treatment arms on the basis of these results. However, we show in five case studies that the results incorporate several simple errors that may be putting patients at risk. One theme that emerges is that the most common errors are simple (e.g., row or column offsets); conversely, it is our experience that the most simple errors are common. We then discuss steps we are taking to avoid such errors in our own investigations.

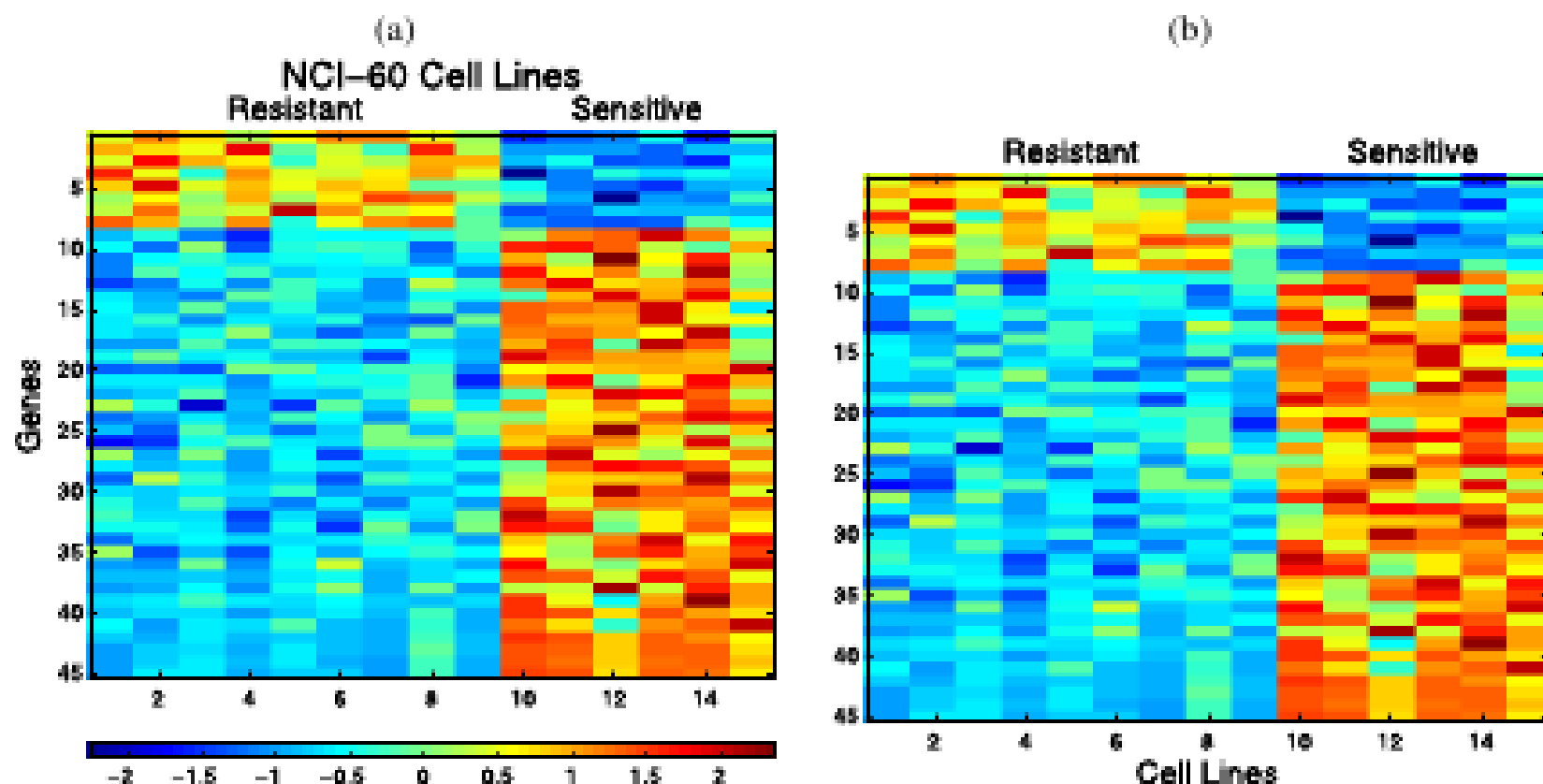
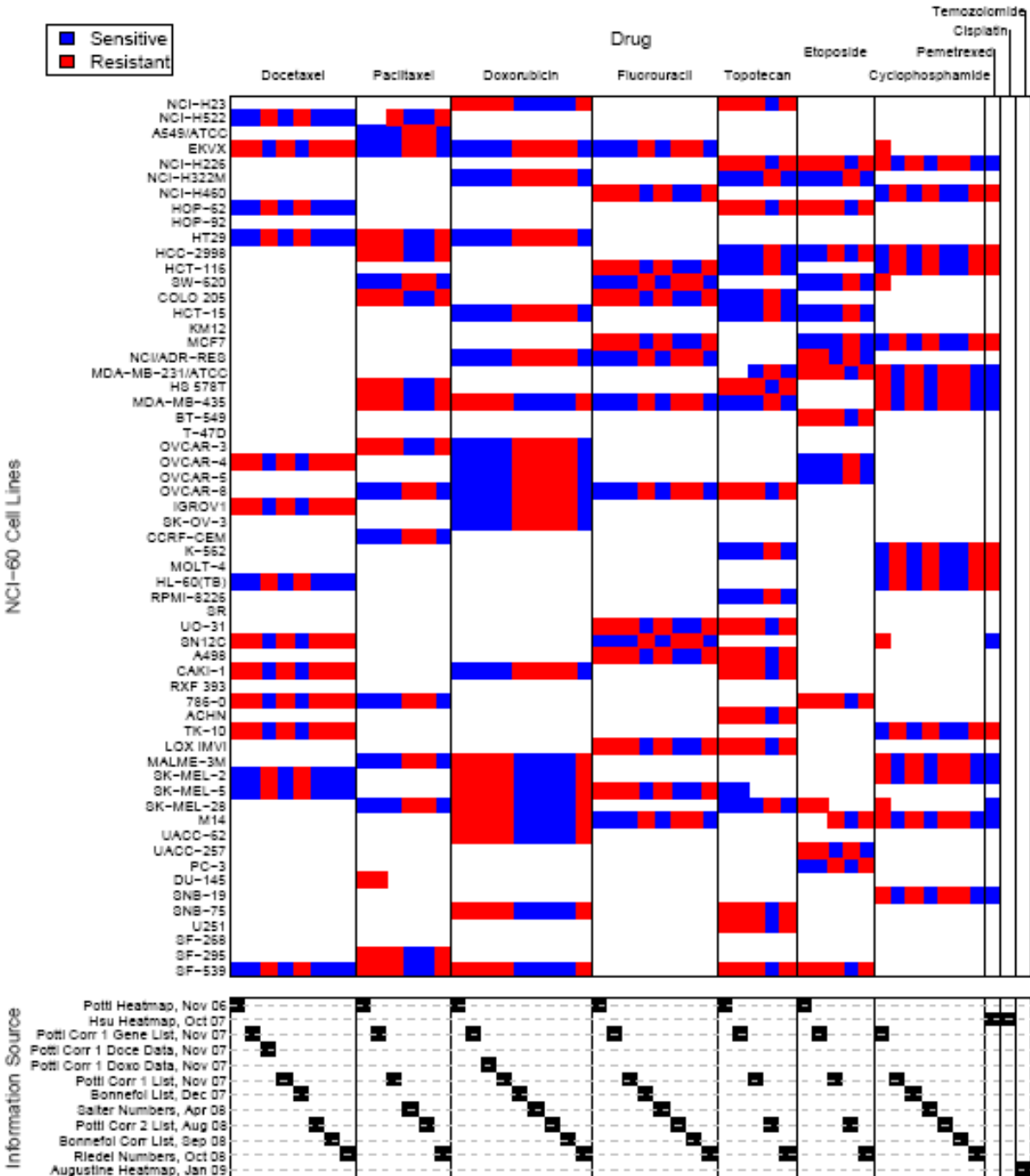


FIG. 4. Approximations to (a) the heatmap initially presented in Figure 4A of Augustine *et al.* (2009) for temozolomide, with lines reportedly chosen from the NCI-60 cell line panel, and (b) the heatmap presented in Figure 1 of Hsu *et al.* (2007) for cisplatin, with cell lines chosen from the 30-line panel of Györfy *et al.* (2006). The heatmaps are the same. We have independently generated the cisplatin heatmap using the Györfy *et al.* (2006) data, but the temozolomide heatmap is neither for temozolomide nor from the NCI-60 panel.

A global overview



Conclusions from Baggerly and Coombes

"Poor documentation led a **report on drug A** to include a **heatmap for drug B** and a **gene list for drug C**.

These results are based on simple visual inspection and counting, and are not documented further."

Corrections made in the journal led to **further problems**.

Conclusions of Baggerly and Coombes

- The most common problems are simple:
 - confounding in the experimental design
 - mixing up the gene labels (off-by-one errors)
 - mixing up the group labels (sensitive/resistant)
- Most of these mix-ups involve simple switches or offsets.
- These mistakes are easy to make, particularly if working with Excel
- ... and/or if working with 0/1 labels instead of names

**We know we will make mistakes.
So...**

1) We should work in a way that will reduce the number of errors

**NO SMOKING
IN LAVATORY**

C1160-03-187



**NO SMOKING
IN LAVATORY**

C1160-03-187



C1160-01-196

"Regardless of whether smoking is allowed in any other part of the airplane, lavatories must have self-contained, removable ashtrays located conspicuously on or near the entry side of each lavatory door "

US Code of Federal Regulations for airworthiness

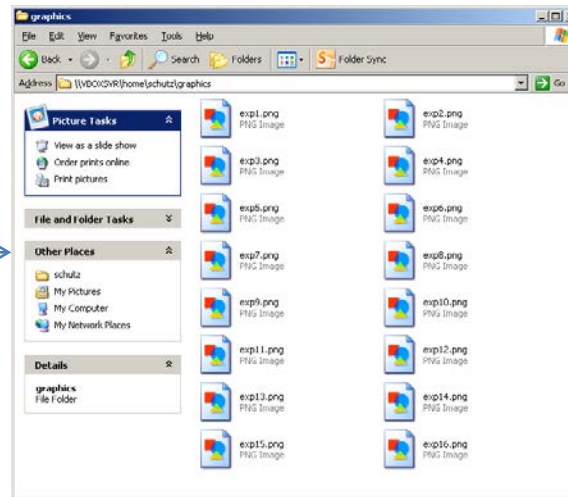
<https://www.law.cornell.edu/cfr/text/14/25.853>



**We know we will still
make mistakes !**

2) We must be able to track what we have done, so we can later find mistakes and correct them

Graphics



Manually

Article/Report

Data analysis

Stats results

H2-D -- CD8

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Nlr5	1	33571	33571	343.424	1.71e-14	***
CIITA	1	77	77	0.785	0.386	
Nlr5:CIITA	1	609	609	6.225	0.021	*
Residuals	21	2053	98			

H2-D -- CD4

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Nlr5	1	31499	31499	90.402	4.65e-09	***
CIITA	1	3467	3467	9.951	0.00478	**
Nlr5:CIITA	1	447	447	1.283	0.27009	
Residuals	21	7317	348			

H2-D -- NK

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
Nlr5	1	18867	18867	70.462	3.78e-08	***
CIITA	1	386	386	1.441	0.243	
Nlr5:CIITA	1	403	403	1.506	0.233	
Residuals	21	5623	268			

Manually

**First steps:
keep track of everything you do,
and all results you get**

- Save your commands in scripts
- When you run a script, save everything it produces (or at least the important results) to a text file.
- Use `sessionInfo()` to record the software used.
- Use GIT to manage the revisions of your files


```
> library(affy)
> sessionInfo()
```

```
[11] LC_MEASUREMENT=en_AU.UTF-8 LC_IDENTIFICATION=C
```

```
attached base packages:
```

```
[1] stats      graphics  grDevices utils      datasets  methods   base
```

```
other attached packages:
```

```
[1] affy_1.34.0      Biobase_2.16.0    BiocGenerics_0.2.0
```

```
loaded via a namespace (and not attached):
```

```
[1] affyio_1.24.0      BiocInstaller_1.4.7  preprocessCore_1.18.0
[4] zlibbioc_1.2.0
```

```
> sink("logfile.txt")
> sessionInfo()
...
> sink()

> capture.output( sessionInfo(), file="logfile.txt" )
> log <- capture.output( sessionInfo() )
[1] "R version 2.15.1 (2012-06-22)"
[2] "Platform: x86_64-pc-linux-gnu (64-bit)"
[3] ""
[4] "locale:"
[5] "  [1] LC_CTYPE=en_AU.UTF-8      LC_NUMERIC=C              "
[6] "  [3] LC_TIME=en_AU.UTF-8      LC_COLLATE=en_AU.UTF-8    "
[7] "  [5] LC_MONETARY=en_AU.UTF-8  LC_MESSAGES=en_AU.UTF-8  "
...
```

Use `sink()` and `capture.output()` if you want to save this information from an R script

"FINAL".doc



FINAL.doc!



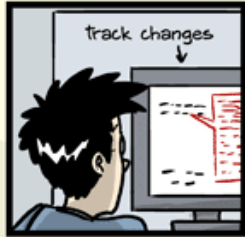
FINAL_rev.2.doc



FINAL_rev.6.COMMENTS.doc



FINAL_rev.8.comments5.
CORRECTIONS.doc



FINAL_rev.18.comments7.
corrections9.MORE.30.doc

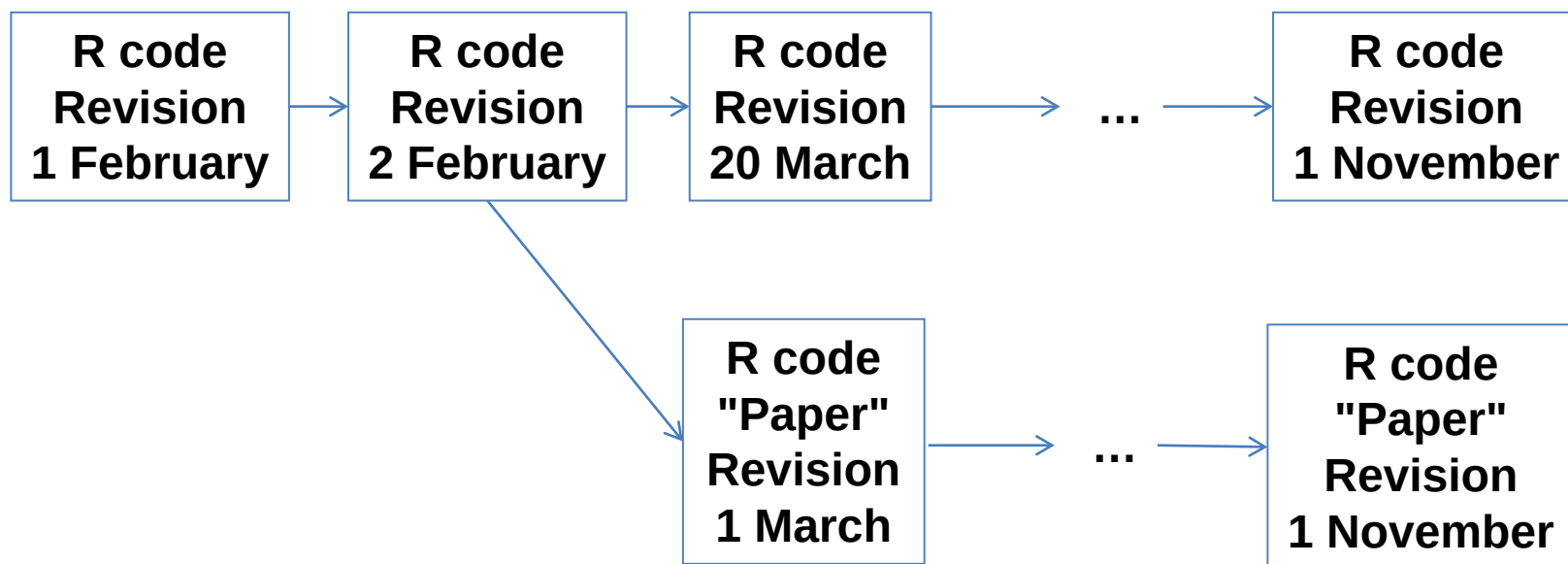


FINAL_rev.22.comments49.
corrections.10. #@\$%WHYDID
ICOMETOGRADSCHOOL?????.doc

JORGE CHAM © 2012

WWW.PHDCOMICS.COM

Use a tool like **GIT** to manage different versions/ revisions/branches of your files.



Dynamic documents with knitr

- Based on the idea of **literate programming** (Donald Knuth, 1984):

Combine program code and explanation/
documentation in same document

- Writing report step by step while processing the data, in the same file
- Documents in which the information is always up-to-date

Allows you to integrate your results in a report.
Write the R code directly with the text, and
later integrate the results directly into the text.

[Home](#)[Objects](#)[Options](#)[Hooks](#)[Patterns](#)[Demos](#)

knitr

Elegant, flexible and fast dynamic report generation with R



Overview

The knitr package was designed to be a transparent engine for dynamic report generation with R, solve some long-standing problems in Sweave, and combine features in other add-on packages into one package (`knitr` $\approx$ `Sweave` + `cacheSweave` + `pgfSweave` + `weaver` + `animation::saveLatex` + `R2HTML::RweaveHTML` + `highlight::HighlightWeaveLatex` + `0.2 * brew` + `0.1 * SweaveListingUtils` + more).

<http://yihui.name/knitr/>

What we need to use knitr

- R
 - `knitr` R package
 - Editor (RStudio)
 - TeX Live (optional, required for creating PDFs)
 - Pandoc
-
- Learn from demos and examples:
 - <http://yihui.name/knitr/>
 - <http://rpubs.com>

RStudio

File Edit Code View Plots Session Build Debug Tools Help

Go to file/function

survey.Rmd x Untitled1* x

Source on Save Run Source

```

1 size <- 100
2 datax <- rnorm(size)
3 class <- sample( c(1,2), size, replace=TRUE)
4 datay <- 2*datax + rnorm(size)
5
6 plot(datax, datay, col=class+1)
7

```

7:1 (Top Level) R Script

Environment History

Global Environment

Values

class	num [1:100]	2 1 2 2 1 2 1 1 1 2 ...
datax	num [1:100]	1.115 -0.294 1.979 0.385 0.399 ...
datay	num [1:100]	2.94 -1.97 4.7 1.28 0.2 ...
size		100

Files Plots Packages Help Viewer

Zoom Export Clear All

Console

Type 'demo()' for some demos, 'help()' for on-line help, or 'help.start()' for an HTML browser interface to help. Type 'q()' to quit R.

```

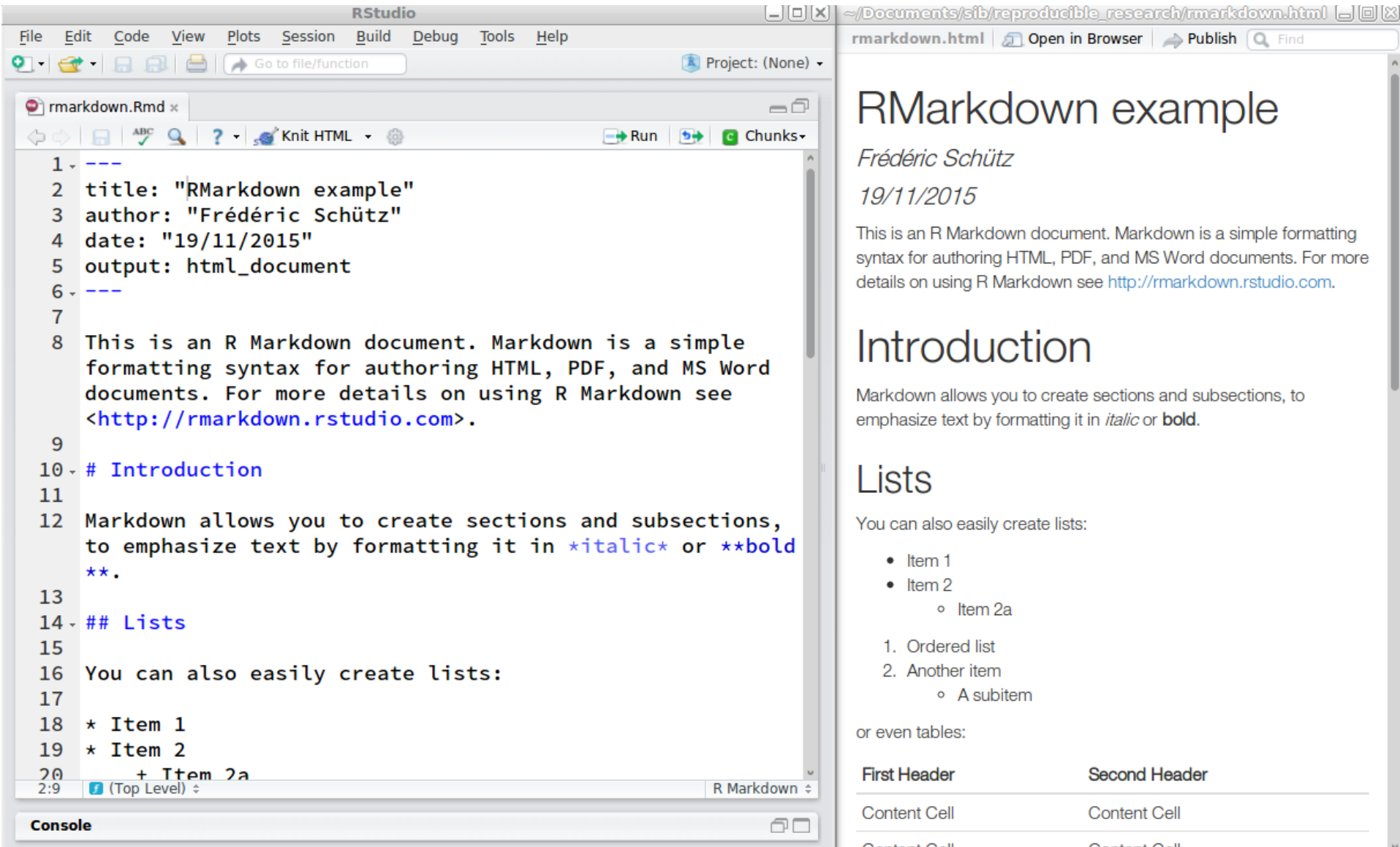
> ?sample
> size <- 100
> datax <- rnorm(size)
> class <- sample( c(1,2), size, replace=TRUE)
> datay <- 2*datax + rnorm(size)
> plot(datax, datay, col=class+1)
> 1

```

Markdown is a simple plain text format that allows you to specify the layout of a document, and which can easily be converted to different formats afterwards.

R Markdown combines the core syntax of markdown (easy-to-write plain text format) with embedded R code chunks that are run so their output can be included in the final document.

R Markdown v2 (<http://rmarkdown.rstudio.com/>)



The screenshot displays the RStudio interface with an R Markdown document open. The left pane shows the source code, and the right pane shows the rendered HTML output.

Source Code (Left Pane):

```
1 ---
2 title: "RMarkdown example"
3 author: "Frédéric Schütz"
4 date: "19/11/2015"
5 output: html_document
6 ---
7
8 This is an R Markdown document. Markdown is a simple
9 formatting syntax for authoring HTML, PDF, and MS Word
10 documents. For more details on using R Markdown see
11 <http://rmarkdown.rstudio.com>.
12
13 # Introduction
14
15 Markdown allows you to create sections and subsections,
16 to emphasize text by formatting it in italic or bold.
17
18 ## Lists
19
20 You can also easily create lists:
21
22 * Item 1
23 * Item 2
24   + Item 2a
```

Rendered Output (Right Pane):

RMarkdown example

Frédéric Schütz

19/11/2015

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

Introduction

Markdown allows you to create sections and subsections, to emphasize text by formatting it in *italic* or **bold**.

Lists

You can also easily create lists:

- Item 1
- Item 2
 - Item 2a

1. Ordered list
2. Another item
 - A subitem

or even tables:

First Header	Second Header
Content Cell	Content Cell
Content Cell	Content Cell

RStudio

File Edit Code View Plots Session Build Debug Tools Help

Go to file/function Project: (None)

rmarkdown.Rmd

Knit HTML Run Chunks

```
33
34
35 # Integrating R code
36
37 When you click the Knit button a document will be
38 generated that includes both content as well as the
39 output of any embedded R code chunks within the
40 document. You can embed an R code chunk like this:
41
42 ```{r}
43 summary(cars)
44 ```
45
46 You can also add R code directly in the text; for
47 example, the dataset cars contains `r nrow(cars)` rows.
48
49 You can also embed plots, for example:
50
51 ```{r, echo=FALSE}
52 plot(cars)
53 ```
54
55 Note that the `echo = FALSE` parameter was added to the
56 code chunk to prevent printing of the R code that
57 generated the plot
```

43:41 (Top Level) R Markdown

Console

~/Documents/sib/reproducible_research/rmarkdown.html

rmarkdown.html Open in Browser Publish Find

Integrating R code

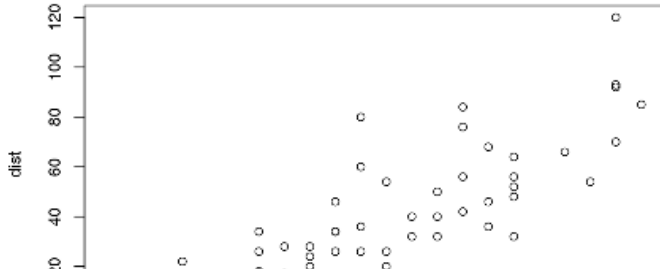
When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
summary(cars)
```

##	speed	dist
## Min.	: 4.0	Min. : 2.00
## 1st Qu.:	12.0	1st Qu.: 26.00
## Median :	15.0	Median : 36.00
## Mean :	15.4	Mean : 42.98
## 3rd Qu.:	19.0	3rd Qu.: 56.00
## Max. :	25.0	Max. : 120.00

You can also add R code directly in the text; for example, the dataset cars contains 50 rows.

You can also embed plots, for example:



A scatter plot showing the relationship between speed (x-axis) and distance (y-axis) for the 'cars' dataset. The x-axis ranges from 0 to 25, and the y-axis ranges from 0 to 120. The plot shows a positive correlation, with distance increasing as speed increases. The data points are represented by open circles.

R Markdown example

```
---  
title: "Untitled"  
author: "Frédéric Schütz"  
date: "23/01/2015"  
output: html_document  
---
```

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see <http://rmarkdown.rstudio.com>.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

R Markdown example (continued)

```
```{r}  
summary(cars)
```
```

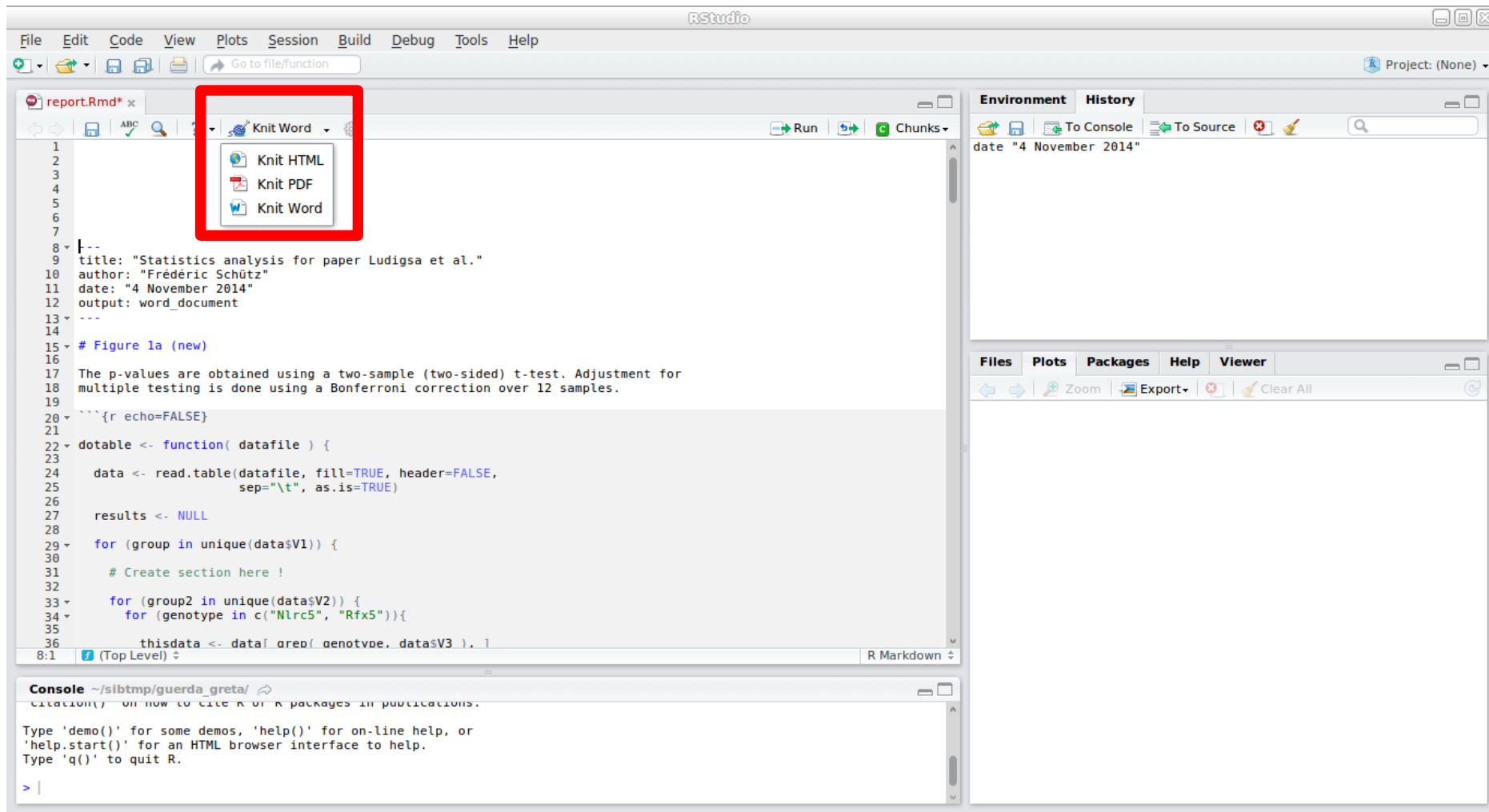
You can also embed plots, for example:

```
```{r, echo=FALSE}  
plot(cars)
```
```

Note that the ``echo = FALSE`` parameter was added to the code chunk to prevent printing of the R code that generated the plot.

- Emphasis: `*italic*` `**bold**`
 `_italic_` `__bold__`
- Headers
Header 1
Header 2
Header 3
- Unordered List:
 - * Item 1
 - * Item 2
 - + Item 2a
 - + Item 2b
- Ordered list:
 - 1. Item 1
 - 2. Item 2
 - 3. Item 3
 - + Item 3a
 - + Item 3b

In R Studio



- R code placed in *chunks* will be evaluated and printed

```
```{r}
summary(cars$dist)
summary(cars$speed)
````
```
- Inline R Code
There were `r nrow(cars)` cars studied
- Links: use a plain http address or add a link to a phrase:
`http://example.com`
`[linked phrase](http://example.com)`
- Images on the web or local files in the same directory:
`![alt text](http://example.com/logo.png)`
`![alt text](figures/img.png)`

Why use knitr ?

- all-in-one: analysis, documenting, formatting, reporting
- no annoying and error-prone copy-pasting
- modifying input data or code: changes are directly reflected in report
- easy to display underlying code in report when needed
- split code in chunks, but can still access all previously defined
- variables (single R session)
- flexible: code externalization, child documents, caching,...

Exercises

- Using Rstudio, start a new .Rmd (R Markdown file).
- Look at the template that was provided, change the R code
- Create an HTML, a Word and a PDF file from this Markdown code
- Make sure you know how to do at least the following: sections, lists, create a table from R, insert a graphic from R.
- Note: you may need to install a TeX distribution to generate PDF; you can also generate a Word or Excel document, and print/convert them to PDF if required
- Make sure to include information about the current R session (R version, packages loaded) in the final document
- Adapt an R script of your choice (ideally one you would use in your work) in a Markdown report